

CompTIA SecAI+

Lab 2.2 — Apply Structured Prompt Templates

Using GPT OSS 20B in Open WebUI

Lab Setup

Structured prompt templates standardise AI outputs, reduce ambiguity, and enforce professional formats in security operations. In this lab you will apply templates to a phishing incident — a security alert summary, an incident triage assessment, and an executive summary — then apply a template to a real alert file.

For the final section, download Alert File1.json and save it in home/kali/Labfiles 2.2.

Before you begin, open Chat Controls (the sliders icon at the top-right of the chat), select Advanced Parameters, and apply these settings:

Parameter	Value
Max Tokens	2048
Every other parameter	Default (do not change)

Note: Do not change num_thread, num_batch, top_k, min_p, or any other advanced parameter — only Max Tokens should be changed. The others are not supported and cause a “property ... is unsupported” error.

Note: The model for this lab is GPT OSS 20B (Direct tab in the model dropdown).

Note: Sections 2.2.1, 2.2.2, and 2.2.3 run in the SAME chat, because the templates all refer back to the INCIDENT CARD you provide in 2.2.1. If you see a “request too large / tokens per minute” message, start a New Chat and paste the INCIDENT CARD again together with the next template prompt.

2.2.1 Security Alert Summary

A common SOC workflow is documenting suspicious activity in a standardised format. The Security Alert Summary template ensures no critical detail is missed and every analyst reports in the same style.

Step 1: Open Firefox and sign into Open WebUI.

Step 2: From the New Chat page, select the model dropdown and choose **GPT OSS 20B**.

Step 3: In the chat, enter the following prompt and note the response:

Prompt 1:

You are a SOC analyst. Read the INCIDENT CARD and write a concise paragraph describing what happened in plain language. Do not add facts not present in the card.

INCIDENT CARD (PHISHING - FINANCE)

- Date (CT): 2025-09-14
- First observed: 09:12:33
- Last observed: 10:05:11
- Targets: Finance staff (~25 inboxes)
- User reports: 3 (two clicked, one entered credentials is unconfirmed)
- Subject (examples): "Payment Advice - URGENT", "Invoice Past Due"
- From: accounts@billingsafe.co (Return-Path: bounce@billingsafe.co)
- Domains: billingsafe.co (registered 2025-09-10)
- Auth results: DMARC=fail, SPF=softfail, DKIM=none
- Links: https://invoice-billingsafe[.]co/login (Microsoft 365 lookalike)
- Attachments: PaymentAdvice.htm (HTML)
- Controls: Initial quarantine applied; gateway rule updated; some emails reached users
- Known impact: None confirmed; credential theft possible but unverified

Insights: Asking an open-ended question gives you a baseline — the model's natural level of detail, structure, and terminology, with something unstructured to compare the template against.

Step 4: In the same chat, enter the following prompt and note the response:

Prompt 2:

Complete the SECURITY ALERT SUMMARY TEMPLATE strictly from the INCIDENT CARD. If a field is unknown, write UNKNOWN.

SECURITY ALERT SUMMARY TEMPLATE

THREAT CLASSIFICATION:

- Severity Level: [CRITICAL/HIGH/MEDIUM/LOW]
- Threat Type: Phishing (Credential Harvesting)
- Confidence Level: [HIGH/MEDIUM/LOW]

TECHNICAL SUMMARY:

- Source: [sender/return-path]
- Target: [teams/departments]
- Attack Vector: Email, Link to fake login
- Affected Services: Email, Authentication

INDICATORS OF COMPROMISE (IOCs):

- Sender Domain(s): [list]
- URL(s): [list]

- Auth Results: [DMARC/SPF/DKIM]
- Attachment(s): [name/type]

TIMELINE:

- First Observed: [time]
- Last Observed: [time]
- Duration: [compute or UNKNOWN]

IMPACT ASSESSMENT:

- Systems Affected: [range or UNKNOWN]
- Data at Risk: Credentials (POSSIBLE)
- Service Disruption: Minimal

RECOMMENDED ACTIONS:

- 1.
- 2.
- 3.

NEXT STEPS:

- 1.
- 2.
- 3.

Insights: This template standardises how security events are documented so every alert includes classification, technical details, indicators of compromise, timeline, impact, and recommended actions — complete, consistent, and actionable regardless of which analyst prepares it.

2.2.2 Incident Triage Template

When multiple alerts come in, analysts need a fast way to sort noise from real threats. The Incident Triage template captures the essential details so incidents can be classified, prioritised, and escalated.

Step 1: In the same chat, enter the following prompt and note the response:

Prompt 1:

```
Complete the INCIDENT TRIAGE ASSESSMENT strictly from the INCIDENT CARD.  
Use the rubric: P2-HIGH if multiple users affected and credential theft is  
plausible but unconfirmed.
```

INCIDENT TRIAGE ASSESSMENT

INCIDENT METADATA:

- Incident ID: INC-2025-0914-FIN-01
- Date/Time Reported: 2025-09-14T10:15:00 CT
- Reporter: SOC queue (user reports escalated)
- Initial Source: Email Security Alerts + User Reports

INCIDENT CLASSIFICATION:

- Category:
- Subcategory:
- Severity: [P1/P2/P3/P4]
- Urgency: [IMMEDIATE/4HR/24HR/48HR]

SCOPE ASSESSMENT:

- Affected Systems:
- Affected Users:
- Affected Services:
- Geographic Scope:

TECHNICAL INDICATORS:

- Attack Vectors:
- Suspicious Domains:
- URLs:
- Attachments:
- Auth Results:

BUSINESS IMPACT:

- Service Availability: Operational
- Data Confidentiality: POTENTIALLY_BREACHED (credentials)
- System Integrity: Intact
- Regulatory Implications: UNKNOWN

Insights: This template is used during the first pass of an investigation to quickly classify an incident, estimate scope, assign severity and urgency, and decide whether it needs escalation — so the most critical threats are handled first.

2.2.3 Executive Summary Template

Executives need information framed in terms of business impact and decisions. The Executive Summary template condenses a complex incident into plain language, highlighting the most critical findings, risks, and recommended actions.

Step 1: In the same chat, enter the following prompt and note the response:

Prompt 1:

```
Produce a CYBERSECURITY EXECUTIVE SUMMARY for leadership, based strictly on the INCIDENT CARD.
```

```
CYBERSECURITY EXECUTIVE SUMMARY
```

```
EXECUTIVE OVERVIEW:
```

```
[2 sentences, plain language, no acronyms.]
```

```
KEY METRICS:
```

- Incidents Detected:
- Incidents Resolved:
- Mean Time to Detection:
- Mean Time to Response:
- Systems Affected:

```
BUSINESS IMPACT:
```

- Immediate:
- Potential:
- Financial:

```
ACTIONS TAKEN:
```

- Quarantined messages and tightened gateway rules
- Blocked sender domains/URLs
- User notifications initiated

```
RECOMMENDATIONS:
```

- 1.
- 2.
- 3.

Insights: Written for executives and senior decision-makers, this template strips away technical jargon and focuses on business impact, risk, and next steps, so leadership can make informed decisions quickly without parsing detailed logs.

2.2.4 Prompt Templates with Internal Data

Now you will apply a structured template to real security data — uploading an alert file and using a template to generate a standardised summary from actual network activity.

Step 1: In Open WebUI, start a New Chat with **GPT OSS 20B**.

Step 2: Beneath the chat box, select **+ More**, then **Upload Files**.

Step 3: Select **Alert File1.json** from the **Labfiles 2.2** folder (/home/kali/Labfiles 2.2), then choose Open.

Step 4: In the chat, enter the following prompt and note the response:

Prompt 1:

```
Analyze the json file, output:  
1) Summary (2 sentences) using info, analysis, threat_level_id, tags.  
2) Key IOCs (max 5 bullets).  
3) Recommendations (3 short bullets).
```

```
If something is missing, write UNKNOWN. Do not add facts.
```

Insights: Uploading a large file straight into the model makes it read the whole dataset at once — slower, and prone to skipping details or inconsistent output. In practice, security teams use Retrieval-Augmented Generation (RAG): files are broken into indexed chunks and only the most relevant pieces are pulled in, which you'll explore in a later lab.

Troubleshooting

Common messages in this lab, and what to do:

Situation	Do this
"Request too large ... tokens per minute (TPM): Limit 8000"	Sections 2.2.1–2.2.3 build up history in one chat. Start a New Chat, paste the INCIDENT CARD again with the next template prompt, and continue. Wait a minute if needed — the limit resets every minute.
The answer stops halfway	The reply hit your Max Tokens value. Usually fine for a filled template; if you need the rest, raise Max Tokens slightly (e.g. 3000) but keep it well under 8000.
"property 'num_batch' / 'top_k' / ... is unsupported"	You customised a parameter that isn't supported. Open Chat Controls → Advanced Parameters and set it back to Default. Only Max Tokens should be changed.